

Passive transport across the phospholipid bilayer of enterocytes is correlated with molecules' lipid-water partition coefficients, but recent findings with wild animals demonstrate that passive intestinal absorption of many molecules is inadequately described by this principle. The general expectations that absorption of hydrosoluble nutrients must be mainly mediated by transporters, and that absorption of hydrosoluble toxins will be nil because transporters are lacking, may be wrong for flying vertebrates. In house sparrows (*Passer domesticus*), the paracellular absorption pathway of hydrosoluble fluorescein has been visualized by confocal laser microscopy. The molecular size and charge selectivity of the pathway is similar to that in typical mammalian models, but the intestine's overall permeability to hydrosoluble compounds is at least three times higher in the birds and bats, relative to nonflying mammals, and the pathway can account for the majority of sugar absorption. The view of liposoluble toxin absorption is also under revision as more is learned about Permeability-glycoprotein (Pgp), a 170 kD membrane bound protein that actively pumps out potentially toxic compounds once they diffuse into the cell membrane. In the woodrat *Neotoma stephensi*, relatively high activity of Pgp summed over the intestine's length is correlated with greater tolerance to toxins in juniper forage, relative to the woodrat's congeners. Pgp levels in Stephen's woodrats may be constitutive, because manipulations of dietary juniper level caused no significant change. Results demonstrate how the intestine can increase passive absorption of hydrosoluble nutrients and mount an effective barrier to permeation by some lipophilic toxins. Supported by Hatch WISO4322 and NSF IBN-0216709.

40.2

METABOLIC ROUTING OF NUTRIENTS IN INSECTS

Diane O'Brien¹: ¹Institute of Arctic Biology, University of Alaska Fairbanks, 902 N. Koyukuk Dr., Fairbanks, AK, 99775-7000.

The process by which animals apportion nutrients among metabolic fates is poorly understood, yet central to life history. Holometabolous insects are an interesting case, as diets and nutrient demands may differ widely between life stages. We have used larval and adult diets that contrast in stable isotope ratios to examine the dynamics of C, N, and amino acids in two insect models, butterflies and fruit flies. In nectar feeding butterflies, essential amino acids are stored from the larval diet and routed into reproduction, whereas nonessential amino acids are synthesized from labile sources of C and N. Whereas C is abundant in the diet, N becomes limiting in species without extensive stores. Recently we have extended this approach to investigate how allocation between reproduction and somatic repair is affected by diet restriction in *Drosophila melanogaster*. In *Drosophila*, fecundity decreases and lifespan is extended by restriction of dietary protein. Although diet restriction extends adult lifespan, somatic turnover (often considered a proxy for repair) is reduced compared to full diet flies. Thus, the importance of investment to somatic repair may depend on reproductive effort. The work was supported by the NSF (IBN-9980344) and The Ellison Medical Foundation. **References:** Min, K.J., M.F. Hogan, M. Tatar, D.M. O'Brien. 2006. "Resource allocation to reproduction and soma in *Drosophila*: stable isotope analysis of carbon from dietary sugar." *Journal of Insect Physiology* 52: (in press). O'Brien, D.M., C.L. Boggs, and M.L. Fogel. 2005. "The amino acids used in reproduction by butterflies: a comparative study of dietary sources using compound specific stable isotope analysis" *Physiological and Biochemical Zoology* 78(5): 819-827.

40.3

POST-ABSORPTIVE FATE OF NUTRIENTS AND THEIR EFFECTS ON METABOLISM AND ANIMAL PERFORMANCE

Christopher Guglielmo¹: ¹Biology, University of Western Ontario, 1151 Richmond St. N, London, Ontario, N6A 5B7, Canada.

Following absorption from the gut, macronutrients enter one of three metabolic pools (carbohydrate, protein, fat). Their fate (storage, oxidation, or biochemical transformation to one of the other pools) is determined by the metabolic state of the animal and by the constraints of available biochemical pathways (e.g. fatty acids cannot be converted to glucose). General patterns of nutrient processing will be reviewed across vertebrates with particular focus on lipids. Long distance migratory birds will then be used as a model system to provide examples of how nutrient availability in the diet and body stores can influence metabolic pathways and animal performance. Using work from my laboratory and others I will discuss: 1) the effects of diet macronutrient composition and body composition on fuel selection during high-intensity endurance exercise, 2) the effects of n-6 and n-3 polyunsaturated fatty acids on muscle membrane phospholipids, exercise performance and peroxisome proliferator activated receptor (PPAR) mediated pathways, 3) preferential mobilization of fatty acids from adipocytes, and 4) protein-mediated transport of fatty acids in muscle. Finally, I will illustrate how measurement of circulating levels of nutrient metabolites can be applied in ecological studies to assess rates of energy gain and loss in free living animals. **McWilliams, S.R., C. Guglielmo, B. Pierce and M. Klaassen.** 2004. Flying, fasting, and feeding in birds during migration: a nutritional and physiological ecology perspective. *J. Avian Bio.* 35:377-393. Reviews lipid metabolism in birds.

40.4

POSTABSORPTIVE FATE OF PLANT SECONDARY COMPOUNDS IN HERBIVOROUS MAMMALS

Denise Dearing¹: ¹Biology, Univ. of Utah, 257 S 1400 E, SLC, UT, 84103.

For mammalian herbivores, ingestion of plant secondary compounds is an unavoidable consequence of consuming plants: thus, herbivores must have physiological mechanisms to process absorbed plant secondary compounds. We have been investigating the postabsorptive metabolism of plant secondary compounds in herbivorous woodrats (*Neotoma*) to elucidate general mechanisms of biotransformation. In addition, we are testing the hypothesis that specialists process plant secondary compounds particular to their diet more efficiently than generalists. To these ends, we have been applying whole organism approaches combined with pharmacological techniques (e.g., enzyme activities, microarrays). Our whole organism studies reveal that the juniper specialist, *Neotoma stephensi*, uses less energy to process juniper than a sympatric generalist, *N. albigula*. The combined results of our pharmacological studies suggest that on a juniper diet, the specialist and generalist differ significantly in the induction and activities of specific P450 enzymes. In particular, the specialist induced high activities of P450 2B on a high juniper diet. Furthermore, the conjugation enzymes differ such that the specialist appeared to employ putatively less costly conjugation reactions (methylation) than the generalist (sulfonation). The results suggest that the juniper specialist has evolved to efficiently process secondary compounds in juniper compared to generalists. Research funded by NSF IBN 0236402.

40.6

DIET, BODY COMPOSITION, AND EXERCISE PERFORMANCE: WHY BIRDS DURING MIGRATION SHOULD BE CAREFUL ABOUT WHAT THEY EAT

Scott McWilliams¹, Barbara Pierce²: ¹Natural Resources Science, University of Rhode Island, 105 Coastal Institute in Kingston, Kingston, RI, 02881, ²Biology, Sacred Heart University, 5151 Park Ave, Fairfield, CT, 06825.

Athletes train and eat to increase their capacity to perform on demand. To fuel high-intensity endurance exercise, mammals use primarily glycogen whereas birds use primarily fatty acid

oxidation. Fatty acid composition of fat stores affects exercise performance in a variety of vertebrates although no such studies focus on birds. We tested the hypothesis that fatty acid composition of migratory birds affects their energy expenditure during exercise. Given that dietary fats primarily determine fat composition of vertebrates, we fed Red-eyed vireos (*Vireo olivaceus*) and European Starlings (*Sturnus vulgaris*) one of two diets that differed only in the amount of monounsaturated (MUFA) and polyunsaturated (PUFA) fatty acids. Blood metabolites confirmed that birds were metabolizing fat during exercise. Resting metabolic rate (2.75 ± 0.32 mlO₂/g/hr) was similar for birds fed either diet. However, during short-term exercise the peak metabolic rate of birds fed more MUFA (28.55 ± 1.47 mlO₂/g/hr) was 25% higher than birds fed more PUFA (21.50 ± 1.76 mlO₂/g/hr). Similarly, during continuous 6-hr flights the energy expenditure of birds fed more MUFA (8.51 ± 0.27 Watts) was 13% higher than birds fed more PUFA (7.5 ± 0.28 Watts). Such whole-animal energetic effects suggest that birds during migration can influence the composition of their body fat by selective feeding and this can affect the energetic cost of flights. Supported by NSF (IBN-9983920) and USDA (RIAES-538748).

41.0: PLENARY LECTURE

41.1

THE COMPARATIVE/ECOLOGICAL PHYSIOLOGY OF NECTAR-FEEDING BIRDS: THE LAST 15 YEARS

Carlos Martinez del Rio¹: ¹Zoology, University of Wyoming, Laramie, WY, 82071-3166.

Nectar-feeding birds have been subjects of research for years. Traditionally, researchers have emphasized their astonishingly high metabolism. Our research stresses the mechanisms that they use to fuel their high metabolism by feeding on nectar. Nectar is a dilute sugar solution, and feeding on it demands peculiar digestive and osmoregulatory characteristics. Hummingbird intestines express 7 to 13 times higher sucrase activity than those of other birds. Sunbirds, a group of Old World nectar-feeding birds, express similarly high levels of intestinal sucrase activity, and therefore are convergent with hummingbirds in this trait. Nectar-feeding birds face an osmoregulatory quandary. When they are feeding, they must get rid of astonishingly high volumes of water. Many nectar-feeding birds are small, have high respiration rates, and have kidneys that cannot concentrate urine. Therefore, when they are fasting, they face dehydration. To solve this quandary, hummingbirds shut-off renal filtration at night and greatly reduce it during the day when they are not feeding. "Adaptive renal failure" may be a trait shared by hummingbirds with other small nectar-feeding birds. Mammals can suffer dire consequences when the kidney stops filtering, which is why we call this event a failure. Hummingbirds, in contrast, experience renal "failure" daily. Hummingbirds have both the guts -and the kidneys, needed to face the challenges of their watery and sugary diets.

43.0: PLENARY LECTURE

43.1

NECROPHYSIOLOGY: BLOOD PRESSURE OF ALL THE FISHES

David Jones¹: ¹Zoology Animal Care, University of British Columbia, 6199 South Campus Road, Vancouver, BC, V6T 1W5, Canada.

Bony fishes have a modified blood vessel between the heart and aorta, the *bulbus arteriosus*, which has unique mechanical properties. Unlike blood vessels which have a J-shaped inflation curve, the bulbus is initially very stiff but soon becomes extremely compliant yielding a steady (plateau) pressure which is the same for all members of a given species. The plateau pressure correlates with mean blood pressure determined *in vivo*. Hence, blood pressure in the living animal can be determined from recordings made after it is dead. Recording physiology of the living animal after death defines "necrophysiology" which was virtually the only approach to circulatory physiology before the advent of the scientific revolution and William Harvey's magnum opus on the circulation of the blood published in 1628. Necrophysiology has had its successes in the past, such as the Arab scholar Ibnul-Nafies' (1208-1288) discovery of the lung circulation and now, three quarters of a Millennium later, in determining blood pressure of the world's fishes. Fortunately, or perhaps unfortunately, access to a wide range of species does not present a problem in view of widespread ocean and artisanal fisheries, the latter targeting a specific ecological niche. We have taken advantage of the necrophysiological approach to study evolutionary physiology of fishes. Blood pressure ranges in fishes from 8 in the seahorse to 120 mmHg in the marlin. We have looked at the influence of body mass, trophic level and warm body parts on blood pressure in fishes using phylogenetically independent contrasts in scombrids, rock and reef fishes. Size and aggression really matter. Supported by NSERC.

LATE ABSTRACT

ACTIVATING MUTATIONS IN THE *ABCC8* GENE IN NEONATAL DIABETES MELLITUS

A. P. Babenko, M. Polak, H. Cavé, K. Busiah, P. Czernichow, R. Scharfmann, J. Bryan, L. Aguilar-Bryan, M. Vaxillaire, and P. Froguel.

Departments of Molecular and Cellular Biology and Medicine, Baylor College of Medicine, Houston; the Faculty of Medicine, René Descartes University, INSERM Unité 0363, Hôpital Necker Enfants Malades, Paris; the Departments of Genetic Biochemistry and Pediatric Endocrinology, Hôpital Robert Debré, Paris; Centre National de la Recherche Scientifique Unité 8090, the Pasteur Institute, Lille, France; and the Department of Genomic Medicine, Imperial College London, Hammersmith Hospital, London.

BACKGROUND: The ATP-sensitive potassium (K⁺ATP) channel, composed of the beta-cell proteins sulfonylurea receptor (SUR1) and inward-rectifying potassium channel subunit Kir6.2, is a key regulator of insulin release. It is inhibited by the binding of adenine nucleotides to subunit Kir6.2, which closes the channel, and activated by nucleotide binding or hydrolysis on SUR1, which opens the channel. The balance of these opposing actions determines the low open-channel probability, P_o, which controls the excitability of pancreatic beta cells. We hypothesized that activating mutations in *ABCC8*, which encode SUR1, cause neonatal diabetes. **METHODS:** We screened the 39 exons of *ABCC8* in 34 patients with permanent or transient neonatal diabetes of unknown origin. We assayed the electrophysiologic activity of mutant and wild-type K⁺ATP channels. **RESULTS:** We identified seven missense mutations in nine patients. Four mutations were familial and showed vertical transmission with neonatal and adult-onset diabetes; the remaining mutations were not transmitted and not found in more than 300 patients without diabetes or with early-onset diabetes of similar genetic background. Mutant channels in intact cells and in physiologic concentrations of magnesium ATP had a markedly higher P_o than did wild-type channels. These overactive channels remained sensitive to sulfonylurea, and treatment with sulfonylureas resulted in euglycemia. **CONCLUSIONS:** Dominant mutations in *ABCC8* accounted for 12 percent of cases of neonatal diabetes in the study group. Diabetes results from a newly discovered mechanism whereby the basal magnesium-nucleotide – dependent stimulatory action of SUR1 on the Kir pore is elevated and blockade by sulfonylureas is preserved.